# Caleb Woldemedhine

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## EDUCATION

**University of Denver** Denver, CO
B.S. in Computer Science (AI & ML Concentration) Sept 2023 - Jun 2027
Minors: Mathematics and Statistics
**GPA**: 3.66
***Relevant Coursework:*** Database Organization & Management, Topics in Artificial Intelligence (Machine Learning & LLMs), Applied Math for Data Science and AI, Machine Learning for IoT and Embedded Devices, Linear Algebra, Introduction to Probability

## EXPERIENCE

**Cincinnati Children's Hospital Medical Center** Remote
***Machine Learning Research Intern — Prasath Lab*** Aug 2026–Present
- Developing and evaluating **multi-agent AI frameworks** using LLMs, vision-language models, and specialized sub-agents for medical image analysis, cancer detection, diagnostic reasoning, and verification.
- Conducting research on machine learning for cancer diagnosis and agentic AI in digital pathology, evaluating model architectures, datasets, and methods to improve diagnostic accuracy and reliability.

**Palo Alto Networks** Santa Clara, CA
***Software Engineering Intern*** June 2026 - Present
- Built an LLM visualization service on a Pub/Sub backend that profiles datasets, selects the correct chart, and converts agent outputs into frontend JSON contracts. Created a sub-agent that infers data schemas and generates mock datasets to test 16 chart types and edge cases, achieving 94% chart-selection accuracy while ensuring all outputs could render in the UI.
- Refactored the system into a single-spec chart catalog and migrated the LLM engine to Vertex AI. Built a seven-worker concurrent E2E testing system that automated multi-step Pub/Sub and HTTP flows, along with BigQuery tables and external views used for schema grounding, test validation, and monitoring.

**Frederick S. Pardee Institute for International Futures** Denver, CO
***Data Technician*** Sept 2025 - Aug 2026
- Cleaned and preprocessed international education datasets using Python and pandas, standardizing country-year coverage, resolving missing or inconsistent values, and running quality checks to prepare tables for database imports and website updates.
- Built a Python import script to convert cleaned Excel datasets into SQL-ready inserts, automate formatting and loading workflows, and verify imported values against the Data Dictionary for accurate database integration.

**Tempus Automation** Highlands Ranch, CO
***Software Engineering Intern*** Jun 2025 - Aug 2025
- Built and evaluated a C# ML.NET model using a custom dataset of computer-vision-derived deck dimensions, material specifications, pricing data, and design constraints to predict material quantities and total project costs, then compared its predictions against expected results from test records.
- Designed a MySQL database for inventory, pricing, and training data, then integrated the ML.NET model into a C# backend that queried the database and sent material recommendations and cost estimates to the frontend as JSON responses.

**Denver Center of African Art** Denver, CO
***Web Development Intern*** Jun 2024 - Sept 2024
- Built a responsive website using HTML, CSS, and JavaScript and integrated Stripe donation payments with receipts and donor record tracking.

## PROJECTS

**Stock Prediction and Analysis Tool**
- Built a stock analysis and forecasting website with a Python backend and JavaScript frontend that gives users an interactive dashboard with price forecasts, sentiment insights, trading signals, and model accuracy results powered by machine learning.

**DuDraw AI Coder**
- Built a full-stack AI website where users enter a prompt and a large language model generates DuDraw code that turns their ideas into interactive drawings and simple games.

## TECHNICAL SKILLS

**Languages & Frameworks**: Python, Java, C, C++, C#, SQL, JavaScript, TypeScript, HTML, CSS, React, Next.js, R, Haskell
**Libraries & Technologies**: PyTorch, TensorFlow, Keras, Scikit-learn, Prophet, Pandas, Plotly, NumPy, Excel, Tableau, REST APIs, AWS, ML.NET, Docker, Git, Linux, JSON

## LEADERSHIP

**National Society of Black Engineers** Denver, CO
***President*** Jun 2025 - Present
- Raised $40,000 to send 35 students to the NSBE National Convention and organized a multi-company hiring fair, leading employer outreach, sponsorships, and event operations to expand student career opportunities.